SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: Artificial Intelligence **COURSE CODE: CSA17**

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively.*(BL3)*

Assessment Tool Weightage: 50%

QUESTIONS: 3

Duration : 1 Hour
Total Marks:30

Annexure A

<u>Constraint-Based Problem Solving</u>

Code of Conduct:

I R DINESH KUMAR (192312258) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Dinesh kumar R

Annexure A

Constraint-Based Problem Solving

1. A hospital needs to assign 3 doctors (D1, D2, D3) to 3 shifts (Morning, Afternoon, Night) such that: Constraints:

   i. D1 cannot work Night shift ii. D2 must work

   before D3 (Morning < Afternoon < Night) iii. Only one

   doctor per shift iv. D3 cannot work Morning

   v. All doctors must be assigned exactly one shift

   Apply Backtracking Search to find a valid assignment with all intermediate steps and constraint checks. State the final valid schedule

   ANSWER:

   Backtracking Steps:
   1. Assign D1 → Morning ✓
   2. Assign D2 → Afternoon ✓
   3. Assign D3 → Night ✓

   Constraint Verification:
   • D1 ≠ Night ✓
   • D2 before D3 ✓
   • D3 ≠ Morning ✓
   • One doctor per shift ✓

   Final Schedule:
   Morning : D1
   Afternoon: D2
   Night : D3

2. A robot operates in a grid environment (5×5). It must move from Start (S) to Goal (G). Obstacles block certain paths.

   Constraints:

   i. Movement allowed: Up, Down, Left, Right

   ii. Cost of each move = 1 iii. Cannot pass through obstacles iv. Use Manhattan Distance heuristic

Using above constraints and BFS Search Algorithm Show step-by-step node expansion and priority queue updates and determine the optimal path and cost**.**

Queue Example:
[S]
[A, B]
[B, C, D]
[C, D, E]

Manhattan Distance:
h(n) = |x1 - x2| + |y1 - y2|

Example:
Robot (1,2), Goal (4,4)
h = |4-1| + |4-2| = 5

Optimal Path Example:
S → A → C → F → G

Total Cost = Number of moves × 1

3. An autonomous rescue robot is deployed in a disaster-affected building to locate and rescue survivors. The robot operates in a partially observable environment where some paths are blocked, and it must make decisions with limited information.

The building is represented as a grid of rooms. The robot starts at position S and must reach a survivor located at G.

Constraints:

- The robot can move up, down, left, right
- Some cells are blocked (obstacles) and cannot be traversed
- The robot has limited sensing capability (can only observe adjacent cells)
- Each movement has a cost of 1, but entering risky zones has an additional cost of 2
- The robot must minimize total cost while ensuring safe navigation
- The robot must update its knowledge dynamically (online search scenario)

Tasks:

a) Analyze all the given constraints and explain how they affect the robot's decision-making process.

b) Develop a solution approach using an appropriate search strategy (e.g., Uniform Cost Search / Online Search Agent). Clearly explain the steps involved. c) Demonstrate how the robot applies AI concepts such as:

- Intelligent agents
- Rationality
- Environment type

c) Justify why your chosen approach is the most suitable for solving this problem under the given constraints.

a) Constraint Analysis

- Robot moves only Up, Down, Left and Right.
- Obstacles cannot be crossed.
- Limited sensing (adjacent cells only).
- Safe move cost = 1.
- Risky zone cost = 3 (1 + 2 penalty).
- Robot updates knowledge while moving.

b) Uniform Cost Search

1. Start from S.
2. Insert S into priority queue (cost = 0).
3. Expand the node with the lowest cost.
4. Generate valid neighbours.
5. Ignore blocked cells.
6. Safe move cost = previous +1.
7. Risky move cost = previous +3.
8. Repeat until Goal is reached.

c) AI Concepts

Intelligent Agent:

- Senses environment.
- Makes decisions.
- Chooses safest path.
- Rescues survivor.

Rationality:

- Minimizes total cost.
- Avoids risky areas.
- Reaches goal safely.

Environment:

- Partially Observable
- Dynamic
- Discrete
- Single Agent

d) Justification

Uniform Cost Search is suitable because it finds the minimum-cost path, considers risky zones, avoids expensive routes, guarantees an optimal solution, and can be combined with online search to update knowledge dynamically.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	25%	Clearly defines the problem and identifies all relevant constraints accurately	Identifies most constraints with minor gaps	Partial understanding; misses key constraints	Fails to identify constraints correctly
Constraint Analysis	25%	Analyzes constraints effectively and prioritizes them logically	Provides reasonable analysis with some prioritization	Limited analysis; weak prioritization	No meaningful analysis or prioritization
Solution Development	25%	Develops optimal and feasible solutions satisfying all constraints	Develops workable solutions with minor limitations	Solutions partially satisfy constraints	Solutions do not meet constraints
Application of Concepts	15%	Effectively applies appropriate concepts, methods, or tools	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply relevant concepts
Justification	10%	Provides strong justification for decisions with logical reasoning	Provides justification with some clarity	Weak or unclear justification	No justification provided